BLUESTAQ™ | NDSU NORTH DAKOTA STATE UNIVERSITY

STRAIGHT ROWS HACKATHON · NOVEMBER 13–14, 2026

Food security is national security.

One drifting tractor is one farm’s bad morning.
The dependency behind it reaches the whole nation.



Earth: NASA / Apollo 17 crew · December 7, 1972

CLOSE TO HOME

North Dakota feeds the nation.

#1 in the U.S. in 2024 for:

- **Spring wheat**
- **Durum wheat**
- **Canola**
- **Sunflowers**
- **Dry edible beans**
- **Flaxseed**

The challenge fleet works near Casselton, in Cass County.

USDA NASS, North Dakota Agricultural Statistics 2025
Image: NASA JSC · ISS Expedition 37 · Sept. 15, 2013



WHY THIS MATTERS

The modern farm runs on space.

58%

of U.S. corn acres planted with auto-steer guidance

USDA ERS · 2016 survey
Up from 5% in 2001

$1.4T

in U.S. private-sector benefits from GPS

RTI for NIST · 1984–2017
2017 dollars

$1B

estimated loss per day in a GPS outage

Up to 50% higher during planting season

When guidance fails, the planting window does not wait.

USDA Economic Research Service · RTI International for NIST, 2019

Meet Bluestaq.

Defense-grade data infrastructure, proven across national security missions, now for every organization.

2018

Founded in Colorado Springs

First built for the Space Force's Space Domain Awareness mission

300+

data types in the Unified Data Library

Bluestaq is its primary developer

200M+

records pulled from the UDL every day

Up to 70 applications 2024 report to Congress

"The question is the same on a farm as on a mission."

Bluestaq About page · Space Systems Command, Mar. 2024 · Breaking Defense, June 2024 · Quote: Bluestaq /ARQ for agriculture, 2026

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Your speakers and judges.

Dr. Mike Ortiz

Principal Research Scientist
and Engineering Manager

Host · Opening remarks

Dr. Seth Harvey

Co-Founder and
Chief Executive Officer

Speaker · Judge

Shane Sherman

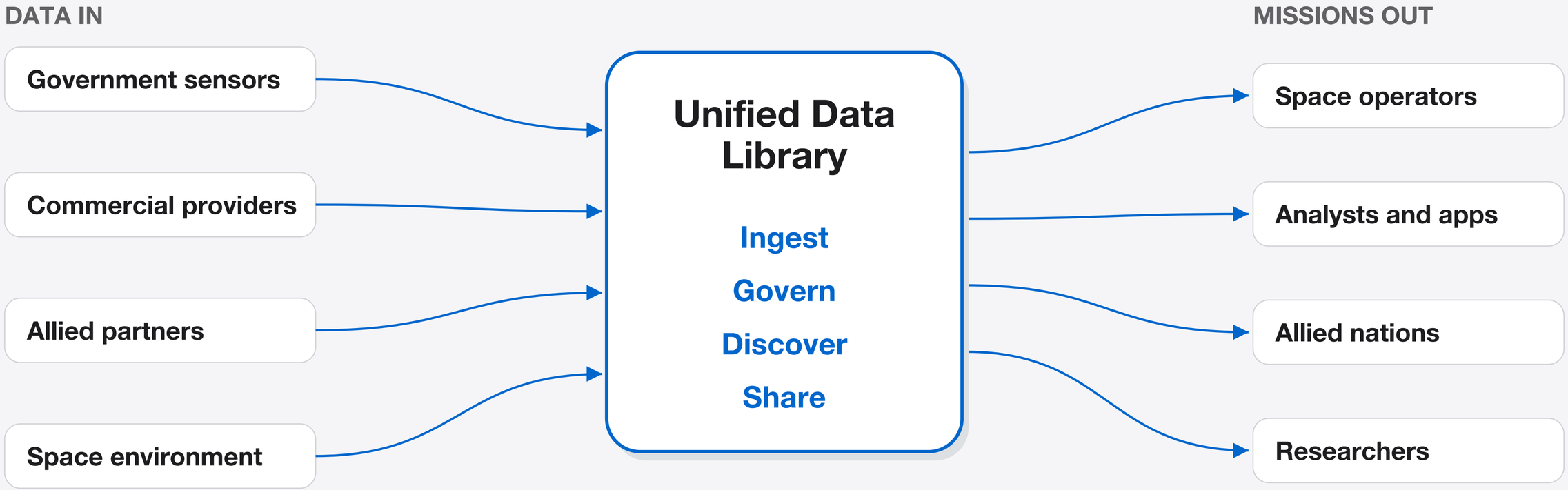
Chief Financial Officer

Speaker · Judge

Judges score every demonstration with the published 100-point rubric.

THE UNIFIED DATA LIBRARY

One library. Many sources. Many missions.



Space Systems Command, Mar. 11, 2024 · Breaking Defense, June 27, 2024 · Bluestaq · Simplified; not every source or user is shown

FROM SIMULATION TO MISSION

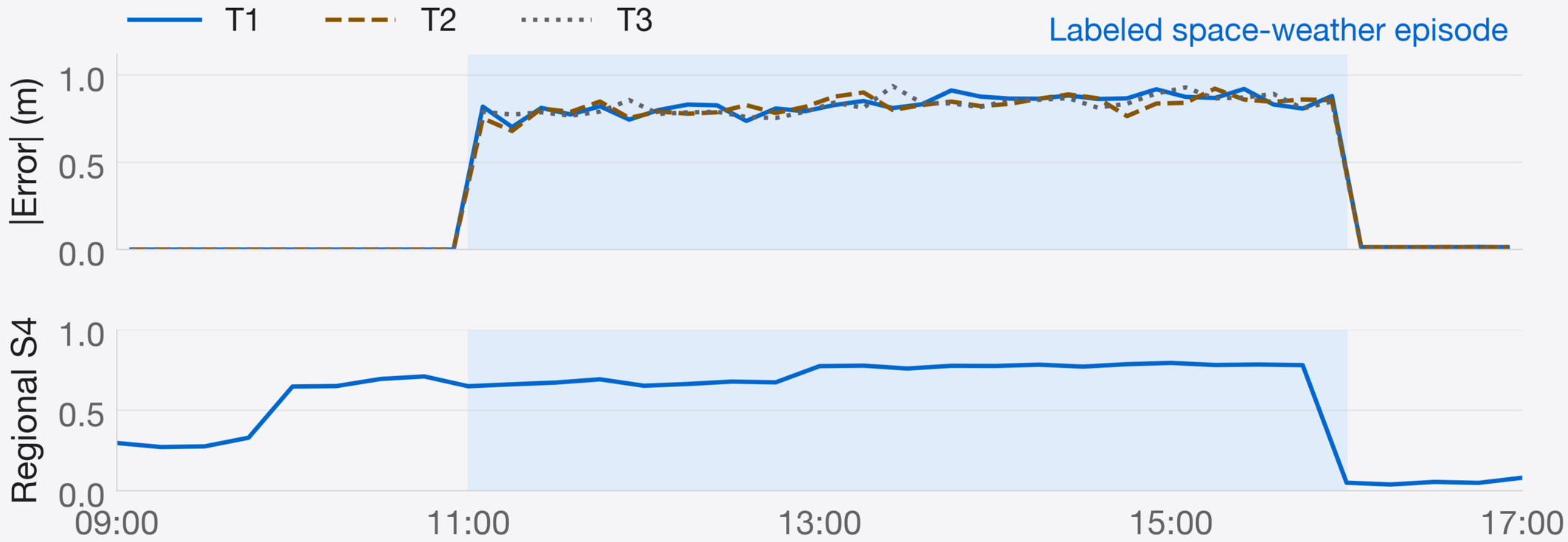
Your challenge, in real-world data.

IN YOUR CHALLENGE		IN THE REAL WORLD
sgi_indices.json Kp and Dst geomagnetic indices	→	**Sgi** Space-weather indices, including Kp and Dst
scintillation_series.json Regional S4 scintillation proxy	→	**IonoObservations** Ionospheric observations · S4 is simplified here
rfi_reports.json Delayed, uncertain interference reports	→	**Emireport** Spectrum-interference reports, including GPS
constellation_advisories.json Notices about satellite status	→	**Notification** Community notices and alerts
tractor_telemetry.csv.gz Satellites, C/N0, fix state, position	→	**GnssObservationset** GNSS receiver observations and status

Different owners, clocks and reliability. Joining them is the job.

Data-type names from Bluestaq’s public UDL SDK (github.com/Bluestaq, Apache-2.0) · All challenge data is simulated

Put the evidence on the same clock.



Public development data · May 5, 2026 (UTC) · Ten-minute medians can hide brief events

THE REPOSITORY

Everything you need, in one place.

bluestaq.github.io/ndsu-hackathon-submissions

Agenda · Get started · Submit · Results · Official guide · Repository: github.com/Bluestaq/ndsu-hackathon-submissions

1 Get the data

AT KICKOFF

Clone the repository, then run

python starter/scripts/get_data.py

The data lands in starter/data/.

2 Build privately

FIRST HOUR

Copy starter/ to your own private repo. Run the baseline, validate, then make one measurable change.

3 Submit one PR

BY SAT 4:00 PM CT

Make your repo public, fork ours, add exactly one file:

submissions/<github-username>/submission.json

GitHub username only. Never your email or student ID. Don’t open your pull request early.

Questions: universityprograms@bluestaq.com

Build your future.

ONE WINNER · BOTH AWARDS

Paid summer internship

Bluestaq HQ
Colorado Springs

Full one-year scholarship

Tuition paid for one year.
Continue your education.

Everything you need: bluestaq.github.io/ndsu-hackathon-submissions

Internship subject to standard eligibility and hiring requirements.

Sources.

USDA · Farm Security is National Security, press release · July 8, 2025
https://www.usda.gov/about-usda/news/press-releases/2025/07/08/farm-security-national-security-trump-administration-takes-bold-action-elevate-american-agriculture

USDA NASS · North Dakota Agricultural Statistics 2025 (2024 data) · Aug. 2025
https://www.nass.usda.gov/sites/default/files/Statistics_by_State/North_Dakota/Publications/Annual_Statistical_Bulletin/2025/ND-Annual-Bulletin25.pdf

USDA ERS · Precision Agriculture in the Digital Era, EIB-248 · Feb. 2023
https://ers.usda.gov/sites/default/files/_laserfiche/publications/105894/EIB-248.pdf

RTI International for NIST · Economic Benefits of GPS · June 2019
https://www.rti.org/publication/economic-benefits-global-positioning-system-gps/fulltext.pdf

Bluestaq · About: history and leadership
https://www.bluestaq.com/us/site/about.html

Bluestaq · /ARQ for agriculture · updated Sept. 9, 2026
https://www.bluestaq.com/ask-assets/content/vert-agriculture.html

Bluestaq release via SpaceNews · primary developer of the UDL · Apr. 8, 2025
https://spacenews.com/bluestaq-expands-globally-with-50m-from-one-bow-river-launches-uk-operations/

Space Systems Command · UDL provides data solutions at the speed of battle · Mar. 11, 2024
https://www.ssc.spaceforce.mil/Newsroom/Article/4039108/space-systems-commands-udl-provides-data-solutions-at-the-speed-of-battle

Breaking Defense · Calvelli report to Congress on the UDL · June 27, 2024
https://breakingdefense.com/2024/06/calvelli-details-plans-to-better-integrate-unified-data-library-into-space-force-ops/

Bluestaq · UDL Python SDK, public API reference (Apache-2.0)
https://github.com/Bluestaq/udl-python-sdk/blob/main/api.md